

ELE 548: Branch Predictors

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1 Tournament Branch Predictor

1.1 Design

1.1.1 Prediction

The tournament branch predictor consists of three components:

1. GAg predictor. The global branch history is mapped to a saturating counter via a table-lookup, and used to make a prediction.
2. PAg predictor. The program counter is mapped to a local branch history via a table-lookup. Then, the local branch history is mapped to a saturating counter via another table-lookup, and used to make a prediction. The second table is shared between all local branch histories.
3. Bimodal choice predictor. The program counter is mapped to a saturating counter via a table-lookup, and used to predict whether the GAg or PAg prediction should be used.

The size of the four tables, the number of bits in the global and local branch histories, and the number of bits in the saturating counters are left as compile-time parameters. The saturating counters are always initialized to weakly-taken. This implies that GAg is the default predictor during warmup.

1.1.2 Update

Each of the three components are updated independently:

1. GAg predictor. The global branch history is extended with the actual branch result. The saturating counter is increased if the branch was taken, and decreased otherwise.
2. PAg predictor. The local branch history is extended with the actual branch result. The saturating counter is increased if the branch was taken, and decreased otherwise.
3. Bimodal choice predictor. The update only occurs if the GAg and PAg predictions disagreed. In this case, the saturating counter is increased if GAg was correct and decreased if PAg was correct.

1.2 Hyperparameter Sweep

All combinations of compile-time parameters detailed in Table 1 were considered. Hyperparameter combinations that exceeded the strict hardware budget of 64KiB (2^{19} bits) were rejected. Likewise, hyperparameter combinations that over-allocated the global or local saturating counter tables (i.e. when the history length is insufficient to index every entry), or did not come within 10% of the hardware budget were rejected. The remaining 118 hyperparameter combinations were simulated.

Hyperparameter	Values
Global history register bits	{17, 18}
Global saturating counter table size	$\{2^{17}, 2^{18}\}$
Global saturating counter bits	{2, 3}
Local history table size	$\{2^{10}, 2^{12}, 2^{14}\}$
Local history bits	{10, 12, 14}
Local saturating counter table size	$\{2^{10}, 2^{12}, 2^{14}\}$
Local saturating counter bits	{2, 3}
Choice saturating counter table size	$\{2^{10}, 2^{12}, 2^{14}\}$
Choice saturating counter bits	{2, 3}

Table 1: Hyperparameter sweep for the tournament branch predictor.

Due to the large search space, I only report MPKIs for the top three hyperparameter combinations. The results are provided in Table 8 and Table 4. The best combinations attained an (approximate) 4.5 MKPI arithmetic mean (3.0 MKPI geometric mean).

1.2.1 Gshare

The hyperparameter sweep detailed in Section 1.2 was repeated using Gshare instead of GAg. Again, due to the large search space, I only report MPKI for the top three hyperparameter combinations. The results are provided in Table 9 and Table 5. The best combinations attained an (approximate) 4.1 MKPI arithmetic mean (2.7 MKPI geometric mean). Gshare clearly outperformed GAg with a 4.1 vs. 4.5 arithmetic mean MKPI.

2 Hashed Perceptron Branch Predictor

2.1 Design

The hashed perceptron branch predictor implementation follows the provided ChampSim¹ implementation without dynamic update thresholds. For simula-

¹<https://github.com/ChampSim/ChampSim>

tion purposes, the XXH32 hash function was used for all reductions. In real hardware, it should be replaced with an efficient reduction scheme.

2.1.1 Prediction

During prediction, a series of perceptron weight tables are indexed with a global history combined with the current program counter, where the length of each global history is geometrically increasing. The corresponding perceptron weights are summed, and compared against a static decision threshold.

2.1.2 Update

If the prediction was incorrect, each corresponding perceptron weight is increased, or decreased by one, to drive the perceptron output closer to, or across the decision threshold.

2.2 Hyperparameter Sweep

All combinations of compile-time parameters detailed in Table 2 were considered. Hyperparameter combinations that exceeded the strict hardware budget of 64KiB (2^{19} bits) were rejected. Likewise, hyperparameter combinations that did not come within 10% of the hardware budget were rejected. The remaining 24 hyperparameter combinations were simulated.

Hyperparameter	Values
Threshold	$\{-30, -1, 0, 1\}$
Perceptron weight table size	$\{2^{11}, 2^{12}\}$
Perceptron weight table count	7, 15
Global history geometric rate	1.2, 1.3, 1.4
Global history geometric minimum	3, 4

Table 2: Hyperparameter sweep for the hashed perceptron predictor.

Due to the large search space, I only report MPKIs for the top three hyperparameter combinations. The results are provided in Table 10 and Table 6. The best combinations attained an (approximate) 3.65 MKPI arithmetic mean (1.43 MKPI geometric mean). This is a substantial improvement over tournament branch prediction with PAg and Gshare.

3 Custom Branch Predictor

3.1 Failed Designs

I was unable to significantly outperform the `ChampSim` hashed perceptron implementation. Here are a few highlights from what I tried:

1. I implemented TAGE based on my lecture notes (≈ 10 arithmetic mean MPKI), with valid bits for table entries (≈ 5 arithmetic mean MKPI), with a meta predictor and Gshare fallback (regression), with a loop predictor (regression), and several other tweaks (regression).
2. I implemented TAGE as described in [2] with valid bits for table entries (≈ 3.5 arithmetic mean MPKI), and several other tweaks (regression).
3. I implemented a tournament between hashed perceptron and Gshare (regression).

3.2 Final Design

My final branch predictor is the hashed perceptron predictor with O-GEHL dynamic update thresholds (as in `ChampSim`) and path history for tables with short global history lengths [1].

3.2.1 Prediction

Prediction is the same as hashed perceptron, except for history lengths below the path history maximum history bits, the path history is folded into the table index. The path history is implemented as a sequence of 4-bit hashes of previous program counters.

3.2.2 Update

Update is the same as hashed perceptron, except the perceptron weights are also updated on a weak prediction. Weak predictions are defined by the O-GEHL dynamic update threshold algorithm. The PC is also folded into a 4-bit hash, and added to the path history.

3.3 Hyperparameter Sweep

All combinations of compile-time parameters detailed in Table 3 were considered. Hyperparameter combinations that exceeded the strict hardware budget of 64KiB (2^{19} bits) were rejected. Likewise, hyperparameter combinations that did not come within 10% of the hardware budget were rejected. The remaining 162 hyperparameter combinations were simulated.

Hyperparameter	Values
Threshold	$\{0, 1, 15\}$
Perceptron weight table size	$\{2^{11}, 2^{12}, 2^{13}\}$
Perceptron weight table count	7, 12, 15, 18, 24
Global history geometric rate	1.2, 1.3, 1.4
Global history geometric minimum	3, 4
Update threshold initial	10
Update threshold speed	16, 18, 20
Path history maximum history bits	8, 16, 22

Table 3: Hyperparameter sweep for the custom hashed perceptron predictor.

Due to the large search space, I only report MPKIs for the top three hyperparameter combinations. The results are provided in Table 11 and Table 7. The best combination attained a 2.767 MKPI arithmetic mean (1.044 MKPI geometric mean), a 0.001 MKPI arithmetic mean reduction over hashed perceptron without path history. This is an improvement over the simplified hashed perceptron, but still worse than TAGE-SC-L which achieves ≈ 2.1 MKPI arithmetic mean.

4 Lessons Learned

In the future, I need to:

1. Design reusable abstractions. I had a hard time writing correct predictors and iterating quickly without good data structural abstractions (e.g. saturating counters).
2. Implement a tracing framework. I wasn’t able to design effective tweaks because I did not trace the predictor state and performance (e.g. table conflicts).

5 LLM Disclosure

I successfully used LLMs for rapid debugging, setting up my build system and generating the result tables. They were ineffective for programming and explaining branch predictor algorithms (and of course, even if they were effective, they would have harmed my learning).

References

- [1] Andre Seznec. “Analysis of the O-GEometric History Length Branch Predictor”. In: *International Symposium on Computer Architecture (ISCA)*. 2005.

- [2] Andre Seznec and Pierre Michaud. “A Case for (Partially) Tagged Geometric History Length Branch Prediction”. In: *The Journal of Instruction-Level Parallelism (JILP)* (2006).

A Results

Hyperparameter	Predictor 1	Predictor 2	Predictor 3
Global history register bits	18	18	18
Global saturating counter table size	2^{17}	2^{17}	2^{17}
Global saturating counter bits	2	3	2
Local history table size	2^{14}	2^{12}	2^{14}
Local history bits	12	14	12
Local saturating counter table size	2^{12}	2^{14}	2^{12}
Local saturating counter bits	3	3	3
Choice saturating counter table size	2^{14}	2^{12}	2^{12}
Choice saturating counter bits	3	3	3

Table 4: Hyperparameter settings for the top-three tournament branch predictor parameter combinations.

Hyperparameter	Predictor 1	Predictor 2	Predictor 3
Global history register bits	18	17	18
Global saturating counter table size	2^{17}	2^{17}	2^{17}
Global saturating counter bits	3	3	3
Local history table size	2^{12}	2^{12}	2^{12}
Local history bits	14	14	14
Local saturating counter table size	2^{14}	2^{14}	2^{14}
Local saturating counter bits	3	3	2
Choice saturating counter table size	2^{12}	2^{12}	2^{12}
Choice saturating counter bits	3	3	3

Table 5: Hyperparameter settings for the top-three tournament branch predictor with Gshare parameter combinations.

Hyperparameter	Predictor 1	Predictor 2	Predictor 3
Threshold	1	0	-1
Perceptron weight table size	2^{12}	2^{12}	2^{12}
Global history geometric rate	3	3	3
Global history geometric minimum	1.4	1.4	1.4
Global history table length	15	15	15

Table 6: Hyperparameter settings for the top-three hashed perceptron predictor parameter combinations.

Hyperparameter	Predictor 1	Predictor 2	Predictor 3
Threshold	1	1	1
Perceptron weight table size	2^{12}	2^{12}	2^{12}
Perceptron weight table count	15	15	15
Global history geometric factor	1.4	1.4	1.4
Global history geometric minimum	3	3	3
Update threshold initial	10	10	10
Update threshold speed	16	18	16
Path history maximum history bits	22	16	16

Table 7: Hyperparameter settings for the top-three custom hashed perceptron predictor parameter combinations.

Trace	Predictor 1	Predictor 2	Predictor 3
SHORT-FP-1	3.273	3.323	3.280
SHORT-FP-2	1.087	1.011	1.087
SHORT-FP-3	0.440	0.439	0.439
SHORT-FP-4	0.253	0.253	0.253
SHORT-FP-5	0.179	0.331	0.179
SHORT-INT-1	7.172	6.156	7.180
SHORT-INT-2	8.105	8.588	8.179
SHORT-INT-3	10.665	10.001	10.667
SHORT-INT-4	2.524	2.385	2.526
SHORT-INT-5	0.381	0.384	0.381
SHORT-MM-1	8.290	8.060	8.290
SHORT-MM-2	10.306	9.837	10.314
SHORT-MM-3	1.200	0.547	1.199
SHORT-MM-4	1.761	1.774	1.760
SHORT-MM-5	4.645	4.780	4.680
SHORT-SERV-1	3.121	4.500	3.342
SHORT-SERV-2	3.212	4.691	3.454
SHORT-SERV-3	4.901	5.503	4.964
SHORT-SERV-4	4.116	5.286	4.281
SHORT-SERV-5	4.249	5.457	4.370
LONG-SPEC2K6-00	3.043	3.080	3.067
LONG-SPEC2K6-01	7.840	7.651	7.840
LONG-SPEC2K6-02	5.263	5.032	5.263
LONG-SPEC2K6-03	5.384	5.109	5.384
LONG-SPEC2K6-04	9.958	9.442	9.958
LONG-SPEC2K6-05	5.419	5.104	5.419
LONG-SPEC2K6-06	2.073	1.690	2.073
LONG-SPEC2K6-07	12.594	13.151	12.773
LONG-SPEC2K6-08	1.338	1.255	1.338
LONG-SPEC2K6-09	5.074	4.995	5.074
LONG-SPEC2K6-10	3.408	3.104	3.410
LONG-SPEC2K6-11	0.805	0.764	0.805
LONG-SPEC2K6-12	12.993	11.939	12.993
LONG-SPEC2K6-13	9.815	10.047	9.863
LONG-SPEC2K6-14	1.200	1.200	1.200
LONG-SPEC2K6-15	1.803	1.858	1.807
LONG-SPEC2K6-16	3.701	3.647	3.702
LONG-SPEC2K6-17	4.726	4.331	4.726
LONG-SPEC2K6-18	1.614	1.588	1.614
LONG-SPEC2K6-19	2.249	2.171	2.250
Arithmetic Mean	4.505	4.512	4.535
Geometric Mean	2.981	2.988	3.001

Table 8: Results for top-three tournament branch predictor combinations in MPKI.

Trace	Predictor 1	Predictor 2	Predictor 3
SHORT-FP-1	2.894	2.909	2.359
SHORT-FP-2	0.898	0.902	0.910
SHORT-FP-3	0.431	0.431	0.432
SHORT-FP-4	0.252	0.252	0.252
SHORT-FP-5	0.330	0.330	0.331
SHORT-INT-1	4.925	5.081	5.070
SHORT-INT-2	6.963	6.975	7.184
SHORT-INT-3	10.166	10.302	10.217
SHORT-INT-4	1.911	1.949	1.897
SHORT-INT-5	0.332	0.338	0.331
SHORT-MM-1	7.842	7.841	7.959
SHORT-MM-2	9.786	9.778	9.960
SHORT-MM-3	0.490	0.511	0.506
SHORT-MM-4	1.717	1.711	1.720
SHORT-MM-5	4.472	4.483	4.543
SHORT-SERV-1	2.449	2.491	2.533
SHORT-SERV-2	2.569	2.608	2.666
SHORT-SERV-3	4.653	4.694	4.781
SHORT-SERV-4	3.844	3.786	3.991
SHORT-SERV-5	3.576	3.547	3.701
LONG-SPEC2K6-00	2.949	2.941	3.052
LONG-SPEC2K6-01	7.552	7.550	7.749
LONG-SPEC2K6-02	4.419	4.595	4.475
LONG-SPEC2K6-03	5.090	5.096	5.067
LONG-SPEC2K6-04	9.345	9.351	9.378
LONG-SPEC2K6-05	5.045	5.032	5.110
LONG-SPEC2K6-06	1.782	1.834	1.821
LONG-SPEC2K6-07	12.499	12.425	12.835
LONG-SPEC2K6-08	1.214	1.218	1.228
LONG-SPEC2K6-09	4.993	4.987	4.903
LONG-SPEC2K6-10	2.689	2.741	2.714
LONG-SPEC2K6-11	0.764	0.759	0.778
LONG-SPEC2K6-12	11.991	11.947	12.013
LONG-SPEC2K6-13	9.268	9.648	9.386
LONG-SPEC2K6-14	1.200	1.200	1.200
LONG-SPEC2K6-15	1.688	1.714	1.687
LONG-SPEC2K6-16	3.638	3.749	3.673
LONG-SPEC2K6-17	4.407	4.482	4.424
LONG-SPEC2K6-18	1.430	1.429	1.454
LONG-SPEC2K6-19	2.131	2.202	2.159
Arithmetic Mean	4.115	4.145	4.161
Geometric Mean	2.697	2.721	2.720

Table 9: Results for the top-three tournament branch predictor with Gshare parameter combinations in MPKI.

Trace	Predictor 1	Predictor 2	Predictor 3
SHORT-FP-1	1.400	1.405	1.398
SHORT-FP-2	1.061	1.067	1.065
SHORT-FP-3	0.026	0.029	0.032
SHORT-FP-4	0.174	0.175	0.176
SHORT-FP-5	0.049	0.050	0.050
SHORT-INT-1	0.434	0.436	0.439
SHORT-INT-2	6.398	6.399	6.403
SHORT-INT-3	10.098	10.097	10.085
SHORT-INT-4	0.893	0.892	0.894
SHORT-INT-5	0.520	0.521	0.522
SHORT-MM-1	10.121	10.127	10.128
SHORT-MM-2	11.402	11.429	11.419
SHORT-MM-3	0.119	0.120	0.124
SHORT-MM-4	1.909	1.912	1.916
SHORT-MM-5	5.297	5.303	5.310
SHORT-SERV-1	1.867	1.873	1.876
SHORT-SERV-2	1.830	1.835	1.839
SHORT-SERV-3	4.823	4.822	4.827
SHORT-SERV-4	3.600	3.605	3.612
SHORT-SERV-5	3.076	3.080	3.085
LONG-SPEC2K6-00	2.547	2.547	2.543
LONG-SPEC2K6-01	8.708	8.707	8.706
LONG-SPEC2K6-02	0.939	0.937	0.939
LONG-SPEC2K6-03	0.901	0.899	0.901
LONG-SPEC2K6-04	10.826	10.817	10.819
LONG-SPEC2K6-05	6.270	6.273	6.273
LONG-SPEC2K6-06	0.988	0.989	0.989
LONG-SPEC2K6-07	10.928	10.923	10.924
LONG-SPEC2K6-08	0.968	0.968	0.968
LONG-SPEC2K6-09	4.650	4.656	4.658
LONG-SPEC2K6-10	0.979	0.980	0.980
LONG-SPEC2K6-11	0.653	0.653	0.653
LONG-SPEC2K6-12	15.265	15.270	15.266
LONG-SPEC2K6-13	7.422	7.419	7.419
LONG-SPEC2K6-14	0.002	0.002	0.002
LONG-SPEC2K6-15	0.639	0.639	0.641
LONG-SPEC2K6-16	3.744	3.743	3.747
LONG-SPEC2K6-17	3.068	3.072	3.072
LONG-SPEC2K6-18	0.135	0.135	0.135
LONG-SPEC2K6-19	1.448	1.451	1.452
Arithmetic Mean	3.654	3.656	3.657
Geometric Mean	1.424	1.431	1.436

Table 10: Results for the top-three hashed perceptron predictor parameter combinations in MPKI.

Trace	Predictor 1	Predictor 2	Predictor 3
SHORT-FP-1	1.206	1.207	1.204
SHORT-FP-2	0.749	0.751	0.749
SHORT-FP-3	0.021	0.021	0.021
SHORT-FP-4	0.094	0.098	0.098
SHORT-FP-5	0.038	0.038	0.038
SHORT-INT-1	0.312	0.312	0.315
SHORT-INT-2	4.976	4.969	4.975
SHORT-INT-3	7.654	7.668	7.667
SHORT-INT-4	0.653	0.655	0.656
SHORT-INT-5	0.318	0.319	0.316
SHORT-MM-1	7.045	7.042	7.044
SHORT-MM-2	8.806	8.803	8.799
SHORT-MM-3	0.089	0.089	0.087
SHORT-MM-4	1.270	1.269	1.270
SHORT-MM-5	4.038	4.044	4.046
SHORT-SERV-1	1.294	1.284	1.286
SHORT-SERV-2	1.258	1.259	1.255
SHORT-SERV-3	3.690	3.690	3.699
SHORT-SERV-4	2.741	2.744	2.743
SHORT-SERV-5	2.297	2.296	2.298
LONG-SPEC2K6-00	1.888	1.889	1.887
LONG-SPEC2K6-01	6.836	6.832	6.829
LONG-SPEC2K6-02	0.687	0.688	0.686
LONG-SPEC2K6-03	0.736	0.740	0.737
LONG-SPEC2K6-04	8.274	8.270	8.272
LONG-SPEC2K6-05	4.730	4.730	4.729
LONG-SPEC2K6-06	0.702	0.702	0.701
LONG-SPEC2K6-07	8.686	8.691	8.696
LONG-SPEC2K6-08	0.762	0.761	0.762
LONG-SPEC2K6-09	3.791	3.791	3.790
LONG-SPEC2K6-10	0.764	0.766	0.764
LONG-SPEC2K6-11	0.513	0.512	0.513
LONG-SPEC2K6-12	10.981	10.984	10.982
LONG-SPEC2K6-13	5.639	5.638	5.641
LONG-SPEC2K6-14	0.001	0.001	0.001
LONG-SPEC2K6-15	0.482	0.481	0.482
LONG-SPEC2K6-16	3.086	3.088	3.085
LONG-SPEC2K6-17	2.408	2.406	2.407
LONG-SPEC2K6-18	0.078	0.072	0.071
LONG-SPEC2K6-19	1.099	1.100	1.100
Arithmetic Mean	2.767	2.768	2.768
Geometric Mean	1.046	1.045	1.044

Table 11: Results for the top-three custom hashed perceptron predictor parameter combinations in MPKI.